higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

ELECTRICAL PRINCIPLES AND PRACTICE
NQF LEVEL 2

(12041002)

27 February 2019 (X-Paper)
09:00–12:00

This question paper consists of 6 pages and 1 formula sheet.

NCV

TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Approximate ALL final answers to TWO decimals.
5. Write neatly and legibly.

NC(V)

QUESTION 1

1.1 Time is a scalar quantity.

Define *time*. (2)

1.2 Convert 2,3 cm^3 to m^3. (2)

1.3 ALL quantities are measured in basic SI units which must be written using certain basic rules.

Give THREE of these rules. (3)

1.4 A specific relationship exists among force, mass and acceleration.

State this relationship and use the basic SI units of mass and acceleration to derive the unit of force. (3)

[10]

QUESTION 2

2.1 Atoms consist of electrons orbiting around a nucleus.

Briefly explain the difference between *electrons* and *free electrons*. (2 + 2) (4)

2.2 An electrical circuit is a path or line through which an electrical current flows.

Draw a neat, labelled light-circuit diagram consisting of a single light, a battery and a switch. (3)

2.3 Georg Simon Ohm was a German physicist and mathematician who discovered 'Ohm's law'.

State Ohm's law and express the law in the form of a mathematical equation. (3)

2.4 Electrical supplies could be either alternating current (AC) or direct current (DC).

Give THREE advantages of AC over DC. (3)

2.5 Resistance is the measure of the opposition to current flow in an electrical circuit.

List THREE factors that influence the resistance of a conductor. (3)

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2.6 A car wash uses a high-pressure cleaner driven by a 4 kW electric motor and connected to a 250 V, 50 Hz supply.

Calculate:

2.6.1 The current drawn from the supply

2.6.2 The resistance of the machine

2.6.3 The energy consumed after 1 hour in MJ

(3 × 2) (6)

2.7 The magnetic field around a current-carrying conductor exists in the form of concentric circles and its direction can be determined by using the right-hand grip rule.

State this rule. (3)

[25]

QUESTION 3

3.1 SIX cells, each with an open-circuit voltage of 1,5 volts and an internal resistance of 0,3 ohms, are connected in parallel across a load resistance of 2,45 ohms.

Calculate:

3.1.1 The total emf (1)

3.1.2 The total internal resistance (1)

3.1.3 The total current flowing through the load resistor (3)

3.1.4 The potential difference across the load resistor (2)

3.2 A transformer is used to step up or step down voltage in an AC circuit.

Briefly explain the difference between a *step-up transformer* and a *step-down* transformer with respect to their windings. (2 + 2) (4)

3.3 An electrical contractor must balance the following loads among the THREE phases of an installation:

- 4 × 1 000 W single-phase heaters
- 3 × 100 W single-phase floodlights
- 5 × 100 W single-phase luminaires

Calculate the approximate load supplied by each phase. (5)

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3.4 An auto transformer has a supply voltage of 230 V and a primary current of 6 A. The number of windings on the primary coil is 200 turns and on the secondary coil 60 turns.

Calculate:

3.4.1 The ratio of the turns

3.4.2 The secondary voltage

3.4.3 The secondary current

(3 × 3) (9)

[25]

QUESTION 4

4.1 Indicate how each of the following measuring instruments are inserted in low-voltage circuits:

4.1.1 Frequency meter

4.1.2 Ohmmeter

4.1.3 Ammeter

(3 × 1) (3)

4.2 There are two types of analogue measuring instruments, namely moving-iron instruments and moving-coil instruments.

Briefly explain the principle of operation of a moving-coil instrument. (3)

4.3 An ammeter is an instrument used to measure current in an electrical circuit.

Show by means of a sketch how to connect an ammeter in a low-voltage circuit. (2)

4.4 An electrician must measure the amount of current flowing through a current-carrying conductor.

List THREE precautions to take when performing this task using a clamp meter. (3)

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4.5 The range of instruments used to measure large DC current and voltages can be extended by connecting additional resistors with the instruments.

Name the resistors and state how each is connected with each of the following meters:

4.5.1 Voltmeter

4.5.2 Ammeter

(2 × 2) (4)
[15]

QUESTION 5

5.1 An electrical-circuit diagram is a simplified schematic representation of an electric circuit.

List THREE requirements that should be met when drawing these circuit diagrams. (3)

5.2 Draw a neat, labelled circuit diagram of TWO single-phase socket outlets supplied from ONE circuit breaker. (6)

5.3 An electrician must install THREE single-phase socket outlets and THREE light bulbs in a three-room house.

Compile a parts list with ALL component ratings for this circuit. (6)
[15]

QUESTION 6

6.1 Both insulators and conductors are essential to the operation of electrical circuits.

6.1.1 Differentiate between *electrical conductors* and *insulators*. (2)

6.1.2 Give TWO properties of carbon as an electric conductor. (2)

6.1.3 Give TWO properties of Bakelite as an insulator. (2)

6.2 An earth-leakage unit is used to automatically disconnect an installation from the supply in the event of an earth fault.

Briefly explain the principle of operation of this safety device. (4)
[10]

TOTAL: 100

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FORMULA SHEET

1 $v = \frac{d}{t}$ 2 $\bar{v} = \frac{d}{t}$ 3 $a = \frac{\Delta v}{\Delta t}$

4 $F = m \times a$ 5 $W = m \times g$ 6 $w = F \times s$

7 $\tau = F \times r$ 8 $\rho = \frac{m}{V}$ 9 $P = \frac{F}{A}$

10 $E = V + Ir$ 11 $V = IR$ 12 $P = VI$

13 $P = I^2 R$ 14 $P = \frac{V^2}{R}$ 15 $E = P \times t$

16 $R = \frac{\rho \ell}{A}$ 17 $A = \pi r^2$ 18 $A = \frac{\pi D^2}{4}$

19 $R_t = R_o(1 + \alpha_0 T)$ 20 $t = \frac{1}{f}$ 21 $\beta = \frac{\Phi}{A}$

22 mmf=NI 23 $H = \frac{mmf}{\ell}$ 24 $H = \frac{NI}{\ell}$

25 $F = \beta I \ell$ 26 $\frac{N_1}{N_2} = \frac{V_1}{V_2} = \frac{I_2}{I_1}$ 27 $S = V_1 I_1 = V_2 I_2$

28 $R_T = R_1 + R_2 + R_3$ 29 $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$ 30 $R_T = \frac{R_1 \times R_2}{R_1 + R_2}$

31 $I_T = I_1 = I_2 = I_3$ 32 $I_T = I_1 + I_2 + I_3$ 33 $I_T = \frac{V_T}{R_T} R$

34 $V_T = V_1 + V_2 + V_3$ 35 $V_T = V_1 = V_2 = V_3$ 36 $V_T = I_T R_T$

37 $Q = I^2 Rt$ 38 $R_{sh} = \frac{I_m R_m}{I_{sh}}$ 39 $V_m = I_m R_m$

40 $R_{se} = \frac{V}{I} - R_m$

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